

Harvesting Radio Frequency Energy

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Most people do not realise that there is an abundance of energy all around us, at all times. We are being bombarded with energy waves all day long. Radio and television towers, satellites orbiting the Earth and even mobile phone antennae are constantly transmitting energy. What if there was a way we could harvest the energy that is being transmitted? If it was possible to gather the energy and store it, we could potentially use it to power other circuits. In case of mobile phones, this power could be used to recharge a battery that is constantly being depleted. The potential exists for mobile phones and even more complicated devices like pocket organisers, personal digital assistants (PDAs) and notebook computers.

Energy harvesting is a process by which ambient energy present in the environment is captured and converted into electrical energy. In recent years, it has become a prominent research area in multiple disciplines. Several energy-harvesting schemes have been exploited including solar energy, mechanical energy, radio frequency (RF) energy, thermal energy, electromagnetic energy, biochemical energy, radioactive energy and so on.

Energy harvesting typically operates in milli-watt or even micro-watt power levels. Almost all energy-harvesting schemes require stages of power condi-

tioning and intermediate storage batteries or capacitors that store energy harvested from the environment.

When we refer to RF harvesting, we do not refer to energy sources that have been specifically designed for powering wireless devices. Instead, we talk about the energy we can collect from public services. In cities and very populated areas, there are a large number of RF sources like broadcasting radio and TV stations, mobile phones base stations and wireless networks. It is possible to collect part of their energy and convert it into useful energy.

RF sources

Radio waves, which are part of the electromagnetic spectrum, consist of magnetic and electrical components. These carry information by varying a combination of amplitude, frequency and phase of the wave within a frequency band. On contact with a conductor such as an antenna, electromagnetic (EM) radiation induces electrical current on the conductor's surface, which is known as skin effect.

Communication devices use antennae for transmission and/or reception of data by utilising different frequency spectrums ranging from 10kHz to 30kHz. Maximum theoretical power available for RF energy harvesting is 7.0 μ W and 1.0 μ W for 2.4GHz and 900MHz frequencies, respectively, for a free space distance of 40m. Path loss of signals will be different in environments other than free space. Table I shows different frequency spectra and its special applications.

Different frequency bands have different applications. Fig. 1 shows the electromagnetic spectrum with different applications surrounding us.

Components of an RF energy-harvesting system

Fig. 3 shows the components of an energy-harvesting circuit. Incident RF power is converted into DC power by the voltage

Fig. 1: Electromagnetic spectrum with surrounding applications

